

Mohamed Zayed Ahmed

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Professional Summary

AI Research Engineer with 3+ years building **production-grade multi-tenant AI platforms** and **agentic systems**. Delivered scalable RAG architectures (Qdrant/HelixDB), high-throughput ML pipelines (3× gains), and LLM evaluation frameworks at Sprints AI and Dark Entry. Strong blend of research (T5 SoTA) and production engineering.

Experience

Sprints AI

Cairo, EG

Research Engineer - R&D

Sep 2025 – Present

- Architected multi-tenant AI Learning Companion with production FastAPI, Supabase, Qdrant/HelixDB hybrid RAG, and Redis state management.
- Built precision content-ingestion pipeline and production agent workflows with persona orchestration and curriculum-aware retrieval.
- Implemented neuroscience-inspired memory systems and dual RAG provider adapters with zero-downtime switching (50% fewer DB round-trips).

Dark Entry

Remote

Research Engineer (Contractor)

Jun 2025 – Dec 2025

- Architected high-throughput ingestion platforms (50+ concurrent feeds) achieving 3× throughput with dual Python/Rust engine.
- Reduced manual intervention by 80% and database queries by 70% through self-healing systems and modular FastAPI refactoring.

Freelance Consulting & Contracts

Remote

ML/AI Engineer - Consultant

Sep 2023 – Present

- Architected multi-agent research frameworks (MCP) and autonomous NER pipelines with active learning + responsible AI guardrails.
- Built LLM evaluation system (3× faster) and pharmaceutical optimization frameworks using GP, BART, and CVAE.

Proteinea

Cairo, EG

Deep Learning Intern

Aug 2022 – Nov 2022

- Developed T5 models for antigen-antibody interactions; thesis later surpassed 2022 SoTA on SABDAB dataset.

Projects

Multi-Agent Research System – Google ADK, FastAPI, Gradio

- SequentialAgent pipeline with external API integration (BraveSearch, ArXiv, Wikipedia) and SQLite persistence for autonomous research workflows.
- Designed scalable architecture with modular agents and clear communication patterns for production deployment.

Media Recommendation System – Milvus, PyTorch, LitServe

- Built ETL pipeline + semantic similarity engine using embeddings and vector DB (Milvus) for context-aware recommendations.
- Combined classical content-based filtering with NLP-powered synopsis analysis for improved relevance.

Antigen-Antibody Translation/Generation – PyTorch, T5, AWS

Graduation Research Thesis

- Custom T5 Transformer achieving 10% higher performance than 2022 SoTA on SABDAB using AWS NVIDIA A10G GPUs.

Skills

Programming: Python, Rust, C++, SQL — **AI/ML:** PyTorch, FastAPI, Triton, Hugging Face, RAG, Multi-Agent Systems, LLM Evaluation, Vector DBs (Qdrant, Milvus)

Infra & MLOps: Docker, AWS, GCP, Redis, Prometheus, OpenTelemetry — **Other:** Git, Linux, PostgreSQL, Jupyter

Education

University of Science and Technology – Zewail City

Giza, EG — Jun 2024

Bachelor of Science, Communication and Information Engineering — Concentration: ML, NLP, Computer Vision